Interference of light

A. Problems on interference in thin films

Formulae used

I. For air – film – air case

1) In transmitted system

   Constructive interference

   $2\mu t \cos r = n\lambda$

   Destructive interference

   $2\mu t \cos r = (2n-1)\lambda/2$

2) In reflected system

   Constructive interference

   $2\mu t \cos r = (2n-1)\lambda/2$

   Destructive interference

   $2\mu t \cos r = n\lambda$

Where,

a) $\mu \rightarrow$ refractive index of the film
b) $t \rightarrow$ thickness of the film
c) $r \rightarrow$ angle of refraction

d) n→ order of interference

e) λ→ wavelength of the light

Solved problems

1. White light falls normally on a film of soapy water of thickness $5x10^{-5}$ cm and refractive index 1.33. Which wavelength in the visible region will be reflected most strongly?

Data:

μ=1.33, t=$5x10^{-5}$ cm, i=0 ∴r=0;

Formula: condition for Constructive interference

$$2\mu t\ cosr=(2n-1)\lambda/2$$

Solution:

$$2\mu t\ cosr=(2n-1)\lambda/2$$

$$\lambda=\frac{4\,\mu t}{(2n+1)}$$

$$=\frac{4\,x\,1.33\,x\,5\,x\,10^{-5}}{(2n+1)}$$

$$=\frac{2.66\,x\,10^{-4}}{(2n+1)}$$

For n=0 λ_1=26,660A^0

For n=1 λ_2=8,866A^0

For n=2 λ_3=5,320A^0

For n=3 λ_4=3,800A^0

Here λ_3=5,320A^0 lies in the visible region ranging

From 4000A^0 to 7000A^0.

Ans: hence, most strongly reflected wavelength is 5,320A^0

2. A beam of parallel rays is incident on a film of thickness 4 x 10^{-5} cm and of refractive index 1.5 at an angle of incidence of 40°. Show that the wavelength 7.228 x 10^{-5} cm will be strongly reflected.

Data:

μ=1.5, t=4x10^{-5} cm, i=40

Formula: condition for Constructive interference

$$2\mu t \cos r=(2n-1)\lambda/2$$

condition for Destructive interference

$$2\mu t \cos r = n\lambda$$

Solution: sinr=$\frac{\sin(i)}{\mu}$=$\frac{\sin 40}{1.5}$=0.4285

$\therefore \cos r$=0.9035

Let us assume that

$$2\mu t \cos r = m\frac{\lambda}{2}$$

Where if m =odd, its constructive

if m =even, its destructive

$$m=\frac{4\mu t \cos r}{\lambda}$$

$$=\frac{4 \times 1.5 \times 4 \times 10^{-5} \times 0.9035}{7.228 \times 10^{-5}}$$

m=3

ans: As m is odd, the film appears bright in the reflected light

3. A soap film of refractive index 1.33 and thickness 1.5 x 10^{-4} cm is illuminated by white light incident at an angle of 45°. In the reflected light a dark band is observed for the wavelength 5 x 10^{-5} cm. Calculate the order of the interference band.

   Data:

   μ=1.33, t=1.5x10^{-4} cm, i=45°

   Formula:

   condition for Destructive interference

   $$2\mu t \cos r = n\lambda$$

   Solution: $\sin r=\frac{\sin(i)}{\mu}=\frac{\sin 45}{1.33}=0.5317$

   $$\therefore \cos r = 0.8469$$

   $$n=\frac{2\mu t \cos r}{\lambda}$$

   $$=\frac{2 \times 1.33 \times 1.5 \times 10^{-4} \times 0.8469}{5 \times 10^{-5}}$$

$n=6.758$

this shows the highest possible order of the visible band is 6

ans: hence order of interference band is 6

4. A soap film of refractive index 1.33 is illuminated by the normal incident of a beam of monochromatic light of wavelength 5893 A°. What is the minimum thickness of the film that will appear (i) dark and (ii) bright in the reflected light?

Data:

$\mu=1.33$, $\lambda=5893$ A°, $i=0 \therefore r=0$

Formula: condition for Constructive interference or brightness

$$2\mu t \cos r=(2n-1)\lambda/2$$

condition for Destructive interference or darkness

$$2\mu t \cos r = n\lambda$$

Solution:

For dark film,

$$t=\frac{n\lambda}{2\mu}$$

$$t_{min}=\frac{\lambda}{2\mu} \qquad \text{as } n=1$$

$$t_{min}=\frac{5893\,x\,10^{-8}}{2\,x\,1.33}=2215\ A^0$$

for bright film,

$$t=(2n+1)\frac{\lambda}{4\mu}$$

$$t_{min}=\frac{\lambda}{4\mu} \qquad \text{as } n=1$$

$$t_{min}=\frac{5893\,x\,10^{-10}}{4\,x\,1.33}=1107\ A^0$$

ans: minimum thickness for dark film is 2215 A^0

minimum thickness for bright film is 1107 A^0

5. A plane wave of monochromatic light falls normally on a uniform thin film of oil which covers a glass plate. The wavelength of the source can be varied continuously. Complete destructive interference is obtained only for wavelengths 5000 A° and 7000 A° and for no other wavelengths in between. Find the thickness of the oil layer .Given, refractive index of oil 1.3 and refractive index of glass = 1.5.

Data:

μ_{oil} =1.3, μ_{glass}=1.5, $_{air}$=1,

$$\lambda_1=5000A^0$$

$$\lambda_2=7000A^0$$

i=0 $\therefore$ r=0

Formula: for Destructive interference

Path difference=(2n-1)λ/2

As the additional P.D of $\lambda/2$ is introduced at each of the points A and B on the denser surface, the condition becomes:

$$2\mu_{oil}\ t_{oil}\ \cos r\pm\lambda/2+\lambda/2=(2n\pm1)\lambda/2$$

$$\therefore 2\mu_{oil}\ t_{oil}\ \cos r=(2n\pm1)\lambda/2$$

Solution:

In the formula

$2\mu_{oil}\, t_{oil} \cos r=(2n\pm1)\lambda/2$

$$\lambda=\frac{2}{(2n\pm1)}2\mu_{oil}\, t_{oil}$$

For λ_1=5000A^0,   n=m+1

λ_2=7000A^0,   n=m

$$\lambda_2=\frac{2}{(2n\pm1)}2\mu_{oil}\, t_{oil}$$

$$\lambda_1=\frac{2}{(2n\pm3)}2\mu_{oil}\, t_{oil}$$

$$\frac{\lambda_2}{\lambda_1}=\frac{(2n\pm3)}{(2n\pm1)}$$

$(2n\pm1)\,\lambda_2= (2n\pm3)\,\lambda_1$

$2m(\lambda_2-\lambda_1)=(3\lambda_1-\lambda_2)$

$$m=\frac{(3\ \lambda 1-\ \lambda 2)}{2(\lambda 2-\ \lambda 1)}$$

$$m=\frac{[(3\,x\,5000)-9000]\,x\,10^{-8}}{2(7000-5000)\,x\,10^{-8}}$$

m=2

using this ,it is found that

$$t_{oil}=\frac{(2m+1)\lambda_2}{4\mu_{oil}}$$

$$=\frac{5\,x\,7000\,x\,10^{-8}}{4\,x\,1.3}=6730\ A^o$$

Ans: thickness of oil layer=6730 A^o

6. White light is incident on a soap film (u = 1.33) at an angle 53° and the reflected light shows two consecutive dark bands corresponding to the wavelengths 6100 A° and 6000 A°. Calculate the thickness of the film.

Data:

μ_{film} =1.33,i= 53°

λ_1=6100A^0

λ_2=6000A^0

Formula: in this case,

condition for Destructive interference or darkness

$2\mu t \cos r = n\lambda$

Solution: $\sin r = \frac{\sin(i)}{\mu} = \frac{\sin 53}{1.33} = 0.6$

$\therefore$ r=36.85°

From the formula, it the foud that the values of μ,r And t are given

Hence

$n\lambda = 2\mu t \cos r$ =constant

$n\lambda$ =constant

$n \;\alpha\; \frac{1}{\lambda}$

$\therefore$ if the consecutive bands are of orders n and (n+1) , the lower order n corresponds to λ_1 and the higher order (n+1) corresponds to λ_2

$\therefore 2\mu t \cos r = n\lambda_1$

$2\mu t \cos r = (n+1)_2$

$n\lambda_1 = (n+1)\lambda_2$

$\therefore n = \frac{\lambda 2}{\lambda 2 - \lambda 1} = 60$

$\therefore$ we get thickness,

$$t = \frac{n\lambda_1}{2\mu . cosr} = \frac{60\, x\, 6100\, x\, 10^{-10}}{2\, x\, 1.33\, x cos\, 36.85} = 1.72x10^{-5}cm$$

ans: thickness of the film is $1.72x10^{-5}$cm

unsolved problems

1. White light is incident at an angle 45^0 on a soap film 4 x 10^{-5} cm thick. Find the wavelength of light in the visible spectrum which will be absent in the reflected Light. Given u=1.2 for soap film
   (ans: λ=7755A^0)

2. A light of wavelength 5800 A° is incident on a thin film of glass of R.I. 1.5 such that the angle of refraction in the plate is 60°. Calculate the smallest thickness of the plate which will make it dark by reflection.
   (ans: t_{min}=3920 A^0)

3. Light travelling in air in incident on a thin parallel film at an angle 57°. If the reflected and refracted rays make an angle of 90° with each other find the refractive index of the film.

(ans: μ=1.539)

4. A parallel beam of sodium light (=5890 A°) strikes a film of oil floating on water .When viewed at an angle of 30° from the normal the 8th dark band is seen. If the refractive index of oil is 1.46, find the thickness of the film.
   (ans: t=1.72A^0)

5. A drop of liquid of volume 0.2 cc spreads over the whole surface of a tank of water of area 1 sq. m. forming a thin film. When white light is incident normally on the film a dark band corresponding to the wavelength 5500 A° is seen in the spectrum. Find the refractive index of the liquid.
   (ans: μ=1.375)

6. Find the thickness of the soap film which appear yellow (λ= 5896 A°) in reflection when it is illuminated by white light at an angle of 45°. Given refractive index of the film =1.33.
   (ans: t= 1308 A^0)

7. A parallel beam of light of wavelength 5870 A° is incident on a thin glass plate of refractive index 1.5 such that the angle of reflection into the plate is 60^0. Calculate the smallest thickness of the glass plate which will appear dark by reflection.
   (ans: t=3913 A^0)

8. White light is incident on a film at an angle of 53.13° and the reflected light shows two consecutive dark bands for the wavelengths 6000 A° and 6100 A°.If the thickness of the film is 0.0017 cm calculate its refractive index.

   (ans: μ=1.33)

B. Problems on wedge shape thin films

Formulae used

1.] $\beta = \frac{\lambda}{2\mu\theta}$

Where,

$\beta \rightarrow$ *Fringe width*

$\mu \rightarrow$ Permeability

$\theta \rightarrow$ Angle of wedge

2.] $\beta = \frac{\lambda}{2\theta}$ (for air film)

3.] $\tan_{\alpha} = \frac{t}{l}$

$t \rightarrow$ Thickness

$l \rightarrow$ Length

Solved Examples

Q1.] Fringes of equal thickness are observed in a thin glass wedge of refractive index 1.52. The fringe spacing is 1mm and the wavelength of light is 5893A^0.Calculate the angle of wedge in second of an arc.

Soln:

Given: μ=1.52, λ =$5893\,X\,10^{-8}\ cm$,

$\beta = 1\,mm = 0.1\,cm$

Formula: the fringe spacing is

$$\beta = \frac{\lambda}{2\mu\theta}$$

The angle of wedge in radians is

$$\theta = \frac{\lambda}{2\beta\mu} = \frac{5893\,X\,10^{-8}\,X\,180}{2\,X\,1.52.0.1\,X\pi}\,X\,3600\ \ second$$

θ= 40 seconds of an arc.

Ans: The angle of wegde is 40 seconds.

Q2.] A wedge shaped air film having an angle of 40 sec is illuminated by monochromatic light and fringes are observed vertically through a microscope. The distance measured between two consecutive bright fringes is 0.12cm.Calculate the wavelength of light used

Soln:

Given: $\theta = 40\ \ secs = \frac{40}{3600}\ \ degrees$

$$= \frac{40\,X\ \ \pi}{3600\,X\,180}\,radians, \qquad \beta = 0.12\ \ cm$$

Formula: spacing between the consecutive bright fringes is,

$$\beta = \frac{\lambda}{2\theta} \qquad \text{(for air film)}$$

The wavelength is, $\lambda = 2\beta\theta = 2\,X\,0.12\,X\,\frac{40\,X\;\pi}{3600\,X\,180}$

$=4654 X 10^{-8} cm$

$\lambda = 4654\,A^{0}$

Ans: The wavelength of light used is 4654 A^{0}

Q3.]Two optically plane glass strips of length 10cm are placed one over the other. A thin foil of thickness 0.01mm is introduced between them at one end to form an air film. If the light used has wavelength 5900 A^{0}. Find the width of consecutive bright frings.

Soln:

Given: l=10cm=0.1m, t=0.01x$10^{-3}m$, μ=1, $\lambda = 5.9\,x\,10^{-7}\,m$

β =?

$$\tan\alpha = \frac{t}{l} = \frac{0.01\,x\,10^{-3}}{0.1} = 10^{-4}\,radian$$

$$\beta = \frac{\lambda}{2\mu\alpha} = \frac{5.9\,X\,10^{-7}}{2\,X\,1\,X\,10^{-4}}$$

$=2.95\,X\,10^{-3}\,m$

β =2.95mm.

Ans: The width of consecutive bright fringes is 2.95mm

Q4.] Two planes glass surfaces in contact along one edge are separated at the opposite edge by a thin wire.If 20 interference fringes are observed between these the sodium light of wavelength of 5893A^0 at normal incidence what is the thickness of the wire?

Soln:

Data:

$$\lambda = 5893\, A^0 = 5893\, X\, 10^{-8}\, cm$$

No of fringes seen=20

μ=1

d= diameter of the wire

Formula:

$$\beta = \frac{\lambda}{2\mu\alpha}$$

Calculation: For a very small α,

$$\alpha = \tan\alpha = \frac{d}{x} \wedge x = 20\beta$$

$$d = x\alpha = 20\beta\alpha = 20\beta X \frac{\lambda}{2\mu\beta}$$

$$= \frac{10\lambda}{\mu} = \frac{10\, X\, 5893\, X\, 10^{-8}}{1}$$

d=58930A^0

Ans: The diameter is $58930\, A^0$

Q5.] An air wedge is formed by placing a sheet of foil between the edges of two glass plates 75 mm from their point of contact. When the wedge is illuminated with light of wavelength 5.8 x 10^{-7} m the fringes are 1.30 mm apart.Calculate the thickness of the foil.

Soln:

Number of bright bands in the air wedge = $\frac{75}{1.30}$

=57.7

= mλ.

Change in vertical height from one fringe to the next = $\frac{\lambda}{2}$

Therefore vertical height for 57.7 fringes = $\frac{[5.8\, x\, 10^{-7}\, x\, 57.7]}{2}$

=$1.67x10^{-5}$ m

Ans: The foil thickness is therefore $1.67x10^{-5}$ m.

Q6.] A surface of refractive index 1.5 is to be coated with a film of magnesium fluoride (μ= 1.38) to minimise the reflection. Calculate the minimum thickness of film for normal incidence of light of wavelength 5000 A.

Soln:

Data $\mu = 1.5$

$\mu_1 = 1.38$

λ = 5000 A^0

Formula : t= $\frac{\lambda}{4\mu}$

The minimum thickness of film is

t = $\frac{\lambda}{4\mu} = \frac{5000}{4\, X\, 1.38}$ = 906 A^0

Ans : The minimum thickness of film is 906 A^0

Unsolved Examples

Q1.] Light of wavelength 6000 A falls normally on a thin wedge shaped film of refractive index 1.4, forming fringes that are 2 mm apart. Find the angle of the wedge.

(Ans=θ = 1.07 x 10^{-4} radian)

Q2.] Interference fringes are produced by monochromatic light falling normally on a wedge shaped film of cellophane of refractive index 1.4. If angle of wedge is 20 seconds of an arc and the distance between successive fringes is 0.25 cm, calculate the wavelength of light.

(Ans:λ =6790 A)

Q3.] A glass wedge of angle 0.01 radian is illuminated by a monochromatic light of wavelength 6000A falling normally on it. At What distance from the edge of the wedge will the 10th dark fringe be observed by the reflected light?

(Ans: d=$2 X 10^{-4}$m)

Q4.] Light of wavelength 5500A falls normally on a thin wedge shaped film of refractive index 1.4 forming fringes that are 2.5mm apart .Find the angle of wedge in second

(Ans:16.2secs)

Q5.]Two plane rectangular pieces of glass are in contact at one edge and are separated at the other end 10cm away by a wire to form a wedge shaped film. When the film was illuminated by light of wavelength 6000A,10 fringes were observed per cm. Determine the diameter of wire.

(Ans:d=0.03mm)

C. Problems on newtons rings

Formulae used

1.] $D_n^2 = 4Rn\lambda$

$D_n \rightarrow$ Diameter of n^{th} ring

R $\rightarrow$ Radius of curvature

n $\rightarrow$ Order

$\lambda \rightarrow$ Wavelength of light

2.] $D_{n+p}^2 = 4(n+p)R\lambda$

$D_{n+p} = Diameter \ \ of \ \ (n+p)^{th} \ \ rings$

3.] $\lambda = \frac{D_{n+m}^2 - D_n^2}{4mR}$

Solved Examples

Q1.]In newton ring experiment the diameter of 4th and 12th dark rings are 0.4cm and 0.7cm. Deduce the diameter of 20th ring.

Soln:

n=4, n+p=12

$D_4=0.4\,cm$ $D_{12}=0.7\,m$

$D_n^2=4Rn\ \lambda,$ $D_{n+p}^2=4(n+p)R\lambda$

$$D_{12}^2-D_4^2=4X8X\ \lambda XR \quad \ldots\ldots\ldots\ldots\ldots(1)$$

SIMILARLY,

$$D_{20}^2-D_4^2=4X16X\ \lambda XR\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(2)$$

Divide eqn (1) by eqn(2),

$$\frac{D_{12}^2-D_4^2}{D_{20}^2-D_4^2}=\frac{4X8}{4X16}=\frac{1}{2}$$

$$D_{20}^2=2D_{12}^2-D_4^2=0.7^2-0.4^2$$

=0.98-0.16=0.82

Diameter of 20th ring is $=\sqrt{0.82}=0.906\,cm$

Ans: Diameter of 20th ring is $=0.906\,cm$

Q2.]The diameter of 5th dark ring in Newton's rings experiment was found to be 0.42.Determine the diameter of 10th dark ring.

Soln:

$D_n^2=4Rn\ \lambda$

As diameter of 5th dark ring =0.42cm.

Now the diameter of 10th dark ring =?

$$\frac{D_5^2}{D_{10}^2} = \frac{4\,X\,5\,R\ \ \lambda,}{4\,X\,10\,R\ \ \lambda,}$$

$$2\left(D_5^2\right) = D_{10}^2$$

$$D_{10} = \sqrt{2\ \ D_5} = \sqrt{2}\,(0.42)$$

= 0.594cm

Ans : The diameter of 10th dark ring is 0.594cm

Q3.] In a Newton's rings experiment the diameter of the 15th ring was found to be 0.59 cm and that of the 5th ring is 0.336 cm. If the radius of curvature of the lens is 100 cm, find the wave length of the light.

Soln: The given data are

Diameter of Newton's 15th ring (D_{15}) = 0.59 cm = 0.59×10^{-2} m

Diameter of Newton's 5th ring (D_5) = 0.336 cm = 0.336×10^{-2} m

Radius of curvature of lens (R) = 100 cm = 1 m

Wave length of light (λ) = ?

$$= \frac{D_{n+m}^2 - D_n^2}{4\,mR} = \frac{\left(0.59\,X\,10^{-2}\right)^2 - \left(0.336\,X\,10^{-2}\right)^2}{4\,X\,1\,X\,10}$$

$$= \frac{0.235204 \, X \, 10^{-4}}{40}$$

$$= 5880 \, X \, 10^{-10} = 5880 \, A^{0}$$

Ans: The wavelength of light is $5880 \, A^{0}$

Q4.] Light waves of wave length 650 nm and 500 nm produce interference fringes on a screen at a distance of 1 m from a double slit of separation 0.5 mm. Find the least distance of a point from the central maximum where the bright fringe due to both sources coincide.

Sol: Given data are

Wave length of first source (λ_1) = 650 nm

Wave length of second source (λ_2) = 500 nm

Distance of screen (D) = 1 m

Separation between slits ($2d$) = 0.5 mm = 0.5×10^{-3} m

Distance from central maximum where bright fringes due to both sources coincide (x) = ?

Let us consider the n^{th} bright fringe of the first source and the m^{th} bright fringe of the second source coincide at a distance of 'x' from central maximum.

Then,

$X = \frac{n\lambda 1}{2d} D = \frac{m\lambda 2}{2d} D$

or,

$n\lambda_1 = m\lambda_2$

or,

$$\frac{n}{m} = \frac{\lambda}{\lambda 2} = \frac{500}{650} = \frac{10}{13}$$

Ans : 10th bright fringe due to the first source coincides with 13th bright fringe due to second source.

Also,

$x = \frac{n\lambda 1}{2d} D = \frac{10\,X\,650\,X\,10^{-9}\,X\,1}{0.5\,X\,10^{-3}} = 0.013\text{m}$

Ans: The bright fringes of both sources will coincide at a distance of 13 mm from central maximum.

Q5.] Light of wave length 500 nm forms an interference pattern on a screen at a distance of 2 m from the slit. If 100 fringes are formed within a distance of 5 cm on the screen, find the distance between the slits.

Soln:

The given data are

Wave length of light (λ) = 500 nm = 500×10^{-9} m

Distance of screen (*D*) = 2 m

Fringe width (β)=$\frac{5}{100} cm = \frac{50}{100} mm = 0.5\, mm$

$= 0.5\, X\, 10^{-3} m$

Separation between slits(2d)=?

$\beta = \frac{D\lambda}{2d}$

2d=$\frac{\lambda D}{\beta} = \frac{500\, X\, 10^{-9}}{0.5\, X\, 10^{-3}} = 2\, X\, 10^{-3} m = 2\, mm$

Ans: The fringe width is $0.5\, X\, 10^{-3} m$ and separation between slit is $2\, mm$

Q6.] What will be the order of the dark ring which will have double the diameter of the 40th dark ring?

Soln:

Data: $D_n = 2\, D_{40}$

$D_{40}^{2} = 4\, XRX\, 40$Xλ=160Rλ

$D_n^{2} = 2\, D_{40}$

4Rnλ$= 4\, x\, 160\, X\lambda$

n=160

Ans: The order will be 160.

Unsolved Examples

Q1.] Newton's rings are observed in the reflected light of wave length 5900 Å. The diameter of 10^{th} dark ring is 0.5 cm. Find the radius of curvature of the lens used. (Ans=1.059m)

Q2.] Calculate the thickness of air film at the 10^{th} dark ring in a Newton's rings system, viewed normally by a reflected light of wave length 500 nm. The diameter of the 10^{th} dark ring is 2 mm.

(Ans=2.5μm)

Q3.] Two slits separated by a distance of 0.2 mm are illuminated by a monochromatic light of wave length 550nm. Calculate the fringe width on a screen at a distance of 1 m from the slits.

(Ans=2.75mm)

Q4.] In a Newton's ring experiment, the diameter of the 5^{th} ring is 0.30 cm and diameter of the 15^{th} ring is 0.62cm. Find the diameter of the 25^{th} ring

(Ans=0.8239cm)

Q5.] In a Newton's ring experiment the diameter of the 10^{th} dark ring changes from 1.4cm to 1.27cm when a liquid is introduced between the lens and plates.Calculate the refractive index of the liquid

(Ans:μ=1.215)

Q6.] In a Newton's ring experiment the diameter of 5^{th} dark ring is 0.336cm.Find the radius of curvature of the plano-convex lens, if the wavelength of light is 5890A. Also find out the radius of 15^{th} dark ring.

(Ans=0.291cm)

D.Problems on anti and highly reflecting films

Solved:

1. A thin antireflection film (μ_f=1.47) is coated on the surface of a solar cell(μ_s=3.6) to minimize reflection losses. If the average wavelength of the sunlight is taken as 5500A°. What minimum thickness of the film should be deposited on the solar cell surface.

Data:

μ_f=1.47, μ_s=3.6, $_a$=1,

λ=5000A^0

i=0 $\therefore$ r=0

Formula: for Destructive interference

Path difference=(2n-1)λ/2

As the additional P.D of λ/2 is introduced at each of the points on the denser surface, the condition becomes:

$2\mu_f$ t cosr $\pm\lambda/2+ \lambda/2=(2n\pm1)\lambda/2$

$\therefore 2\mu_f$ t cosr=$(2n\pm1)\lambda/2$

Solution:

We have , for r=0;

$2\mu_f$ t = (2n±1)/2

$t= (2n\pm1)\frac{\lambda}{4\mu_f}$

$\therefore \quad t_{min}=\frac{\lambda}{4\mu_f}=\frac{5500\,x\,10^{-10}}{4\,x\,1.47}=935\ A^0$

Ans: minimum thickness is 935 A^0.

2. In costume jewellery, rhinestones(made of the glass with μ_r=1.5) are often coated with silicon monoxide(μ_s=2) to make them more reflective. How thick should be the coating to achieve strong reflection for 560 nm light incident normally?

Data:

μ_r=1.5, μ_s=2, $_a$=1,

λ=560nm

i=0 $\therefore$ r=0

Formula: for strong reflection,

$$p.d = n\lambda$$

Solution: due to reflection at the denser surface,

$$p.d = 2\mu_s\, t \cos r \pm \lambda/2$$

$$\therefore \quad 2\mu_s\, t \pm \lambda/2 = n\lambda$$

$$\therefore \quad 2\mu_s\, t = (2n\pm1)\lambda/2$$

$$t = (2n-1)\frac{\lambda}{2\mu_s} = \frac{560\, x\, 10^{-9}}{4\, x\, 2} = 70nm$$

ans: minimum thickness=70nm

Unsolved

1. A thin transparent film of MgF2 (μ_f = 1.38) is to be coated on a glass microscope lens (μ_g=1.5) to increase the transmission of the normally incident light of wavelength 5893 A°. What minimum thickness should be deposited on the lens surface?
   (t_{min}=1.607A^o)